

Hubble Reconstruction from Acoustic Dictionary Closure: Resolution of the H_0 Tension within LoNalogy

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Related deposits:

LoNalogy framework (v87.1): [doi:10.5281/zenodo.19115338](https://doi.org/10.5281/zenodo.19115338)

Yang–Mills proof (v90): [doi:10.5281/zenodo.19183838](https://doi.org/10.5281/zenodo.19183838)

Particle cosmology (v9): companion paper

Abstract

The so-called Hubble tension—the $\sim 5\sigma$ discrepancy between $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck CMB inference) and $H_0 = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (SH0ES distance ladder)—is shown to be a dictionary-dependent artefact rather than a fundamental physical problem. Within the LoNalogy framework, H_0 is not a primitive quantity but a derived closure relation: $h^2 = \omega_r + \omega_b + \omega_c + \omega_\Lambda$, where all physical densities $\omega_i = \Omega_i h^2$ are fixed by the framework without free parameters.

We prove that the unique no-new-field minimal completion of the early-time sector is a single real cusp-defect scalar δx —the fluctuation of the modulus already present in the four-dimensional LoNalogy action—with decoupling entropy load $g_{*s}^{\text{dec}} = 2^{\text{rk } V} = 2^5 = 32$, giving

$$\Delta N_{\text{eff}}^{\text{hid}} = \frac{4}{7} \left(\frac{10.75}{32} \right)^{4/3} = 0.13345 \dots, \quad N_{\text{eff}}^{\text{LoN}} = 3.17945 \dots$$

No continuous free parameter is introduced. The resulting acoustic scale is

$$100\theta_*^{\text{LoN}} = 1.04123,$$

matching the Planck raw observable $100\theta_*^{\text{obs}} = 1.0411 \pm 0.0003$ to 0.4σ . The reconstructed Hubble parameter is

$$H_0^{\text{LoN}} = 70.71 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

which is neither the CMB inference value nor the local-ladder value, but the unique output of the LoNalogy closure relation.

The 5.8% shift from the base- Λ CDM baseline is shown to originate not from the dark-energy equation of state ($w + 1 \sim Q_{\text{ker}}^2 \sim 10^{-6}$) but from the absolute vacuum amplitude $\Lambda_{\text{eff}} = \Lambda_*^4 Q_{\text{ker}}^{20}$, which exceeds the PDG 2025 value by 17%. The residual 3.4% discrepancy with the SH0ES value corresponds to a 0.074 mag calibration offset in the Cepheid–SN Ia distance ladder, not a contradiction in background expansion.

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Introduction

The Hubble constant H_0 parametrises the present-day expansion rate of the universe. Two classes of measurement have converged on discrepant values: cosmic microwave background (CMB) observations analysed under the base- Λ CDM model yield $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1], while the Cepheid–SN Ia distance ladder gives $H_0 = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [2, 3]. The discrepancy exceeds 5σ and has been termed the “Hubble tension.”

This paper demonstrates that within the LoNalogy framework [8, 6, 7], the Hubble tension is not a fundamental physical problem but a consequence of treating H_0 as a primitive quantity. The LoNalogy framework derives all cosmological densities from internal geometric data—the strict kernel nome Q_{ker} , the electroweak scale $\Lambda_* = 246 \text{ GeV}$, and the broken orbit dimension $\nu_{\text{br}} = 12$ —without free parameters. In this framework, H_0 is a derived quantity: the square root of the sum of physical densities.

The paper proceeds as follows. Part I establishes that H_0 is derived, not primitive, and reconstructs it from closed LoNalogy densities. Part II identifies the unique no-new-field hidden radiation mode and fixes its decoupling entropy load from the rank of the visible fiber bundle. Part III computes the acoustic scale θ_* and demonstrates closure with the Planck raw observable. Part IV addresses the local distance ladder and shows that the residual discrepancy is a calibration offset.

Throughout, we use the LoNalogy conventions and results established in the companion papers [6, 7, 8]. All numerical inputs are:

Symbol	Value	Source
Λ_*	246 GeV	Electroweak VEV
Q_{ker}	1.5677×10^{-3}	Strict kernel nome [6]
ν_{br}	12	(3, 2)-split of SU(5) [6]
N_{fam}	3	Cusps of $X(2)$ [6]

Part I

H_0 as a Derived Quantity

1 The Closure Relation

1.1 Friedmann equation in physical-density form

Definition 1.1 (Physical densities). For each cosmological component $i \in \{r, b, c, \Lambda\}$ (radiation, baryons, cold dark matter, dark energy), define the physical density parameter

$$\omega_i := \Omega_i h^2, \quad (1.1)$$

where $\Omega_i = \rho_i / \rho_{\text{crit}}$ is the density fraction, $h = H_0 / (100 \text{ km s}^{-1} \text{ Mpc}^{-1})$ is the reduced Hubble parameter, and $\rho_{\text{crit}} = 3H_0^2 / (8\pi G_N)$ is the critical density.

Theorem 1.2 (H_0 closure relation). *In a spatially flat Friedmann–Robertson–Walker (FRW) universe, the reduced Hubble parameter satisfies*

$$\boxed{h^2 = \omega_r + \omega_b + \omega_c + \omega_\Lambda.} \quad (1.2)$$

Consequently, H_0 is not an independent parameter but is determined by the four physical densities.

Proof. The Friedmann equation at the present epoch ($z = 0$) for a spatially flat universe ($k = 0$) reads

$$H_0^2 = \frac{8\pi G_N}{3} \left(\rho_r^{(0)} + \rho_b^{(0)} + \rho_c^{(0)} + \rho_\Lambda \right). \quad (1.3)$$

Dividing both sides by H_0^2 gives the constraint

$$1 = \Omega_r + \Omega_b + \Omega_c + \Omega_\Lambda. \quad (1.4)$$

Multiplying both sides by $h^2 = H_0^2/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})^2$:

$$h^2 = \Omega_r h^2 + \Omega_b h^2 + \Omega_c h^2 + \Omega_\Lambda h^2 = \omega_r + \omega_b + \omega_c + \omega_\Lambda. \quad (1.5)$$

Since $h = H_0/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})$, specifying the four physical densities uniquely determines H_0 . $\square$

Remark 1.3. This elementary identity is well known in cosmology. What is new in LoNalogy is that all four ω_i on the right-hand side are *derived from internal geometric data without free parameters*, so h is fully determined.

1.2 LoNalogy values of the physical densities

We now collect the physical densities derived in the companion papers.

Proposition 1.4 (Baryon physical density). *The baryon-to-photon ratio is [7]*

$$\eta_B^{\text{LoN}} = 96 Q_{\text{ker}}^4 \sin \delta_{\text{port}}, \quad (1.6)$$

where $96 = (\dim \mathfrak{su}(5)/N_{\text{fam}}) \cdot \nu_{\text{br}} = 8 \times 12$ and δ_{port} is the portal CP phase. At the minimal closed-form value $\sin \delta_{\text{port}} = 1$ (maximal portal CP):

$$\eta_B^{\text{LoN}} = 96 \times (1.5677 \times 10^{-3})^4 = 5.794 \times 10^{-10}. \quad (1.7)$$

Using the standard BBN conversion $10^{10} \eta_B = 274 \Omega_b h^2$:

$$\boxed{\omega_b^{\text{LoN}} = \frac{10^{10} \times 5.794 \times 10^{-10}}{274} = 0.02116.} \quad (1.8)$$

Proof. The baryon asymmetry derivation is given in full in [7], Theorem 11.1. The BBN conversion factor 274 is the standard relation [4] between η_B and $\Omega_b h^2$ assuming three light neutrino species and standard nuclear rates. Numerically: $Q_{\text{ker}}^4 = (1.5677 \times 10^{-3})^4 = 6.035 \times 10^{-12}$; $96 \times 6.035 \times 10^{-12} = 5.794 \times 10^{-10}$; $5.794/274 = 0.02116$. $\square$

Proposition 1.5 (Dark matter physical density). *With the adjoint-family isotropy (AFI) coupling $g_{\text{hid}}^2 = N_{\text{fam}}/\dim \mathfrak{su}(5) = 3/24 = 1/8$ and $\kappa_j(x_{\text{ker}}) = 1 - x_{\text{ker}}^2 = 0.9964$ [7]:*

$$g_{\text{hid}}\kappa_j = \frac{0.9964}{\sqrt{8}} = 0.3523. \quad (1.9)$$

The s-wave annihilation cross section for $\chi\bar{\chi} \rightarrow A'A'$ in the light-mediator limit is

$$\langle\sigma v\rangle = \frac{(g_{\text{hid}}\kappa_j)^4}{16\pi m_\chi^2} = \frac{0.01541}{16\pi \times (338)^2 \text{ GeV}^{-2}} = 3.13 \times 10^{-26} \text{ cm}^3\text{s}^{-1}. \quad (1.10)$$

The standard Lee–Weinberg freeze-out formula gives

$$\boxed{\omega_c^{\text{LoN}} = 0.120 \times \frac{3.00 \times 10^{-26}}{3.13 \times 10^{-26}} = 0.1150.} \quad (1.11)$$

Proof. The full derivation is in [7], Proposition 10.2. The mass $m_\chi = 338 \text{ GeV}$ is from the heavy-law $M_H = \Lambda_*/k'_{\text{ker}}$ evaluated at the strict kernel. The factor $3.00 \times 10^{-26} \text{ cm}^3\text{s}^{-1}$ is the canonical thermal relic cross section for $\Omega h^2 = 0.120$ [4]. $\square$

Proposition 1.6 (Dark energy physical density). *The effective cosmological constant is [7]*

$$\Lambda_{\text{eff}} = \Lambda_*^4 Q_{\text{ker}}^{20} = 2.944 \times 10^{-47} \text{ GeV}^4. \quad (1.12)$$

Converting to the physical density parameter using the PDG 2025 critical density coefficient $\rho_/h^2 = \rho_{\text{crit}}/h^2 = 1.87834 \times 10^{-29} \text{ g cm}^{-3}$ [4]:*

$$\rho_\Lambda^{\text{LoN}} = \Lambda_{\text{eff}} \times \frac{1}{(\hbar c)^3} = 6.839 \times 10^{-30} \text{ g cm}^{-3}. \quad (1.13)$$

Therefore

$$\boxed{\omega_\Lambda^{\text{LoN}} = \frac{\rho_\Lambda^{\text{LoN}}}{\rho_*/h^2} = \frac{6.839 \times 10^{-30}}{1.87834 \times 10^{-29}} = 0.36372.} \quad (1.14)$$

Proof. The cosmological constant derivation is in [7], Theorem 5.1. Numerically: $\Lambda_*^4 = (246 \text{ GeV})^4 = 3.663 \times 10^9 \text{ GeV}^4$; $Q_{\text{ker}}^{20} = (1.5677 \times 10^{-3})^{20} = 8.037 \times 10^{-57}$; $\Lambda_{\text{eff}} = 3.663 \times 10^9 \times 8.037 \times 10^{-57} = 2.944 \times 10^{-47} \text{ GeV}^4$. The conversion from GeV^4 to g cm^{-3} uses the factor $1 \text{ GeV}^4 = 2.3201 \times 10^{17} \text{ g cm}^{-3}$: $\rho_\Lambda^{\text{LoN}} = 2.944 \times 10^{-47} \times 2.3201 \times 10^{17} = 6.839 \times 10^{-30} \text{ g cm}^{-3}$. $\square$

Remark 1.7. The ratio $\Lambda_{\text{eff}}/\rho_\Lambda^{\text{obs}} = 2.944/2.510 = 1.173$, where $\rho_\Lambda^{\text{obs}} = 2.51 \times 10^{-47} \text{ GeV}^4$ [4], shows that the LoNalogy vacuum energy exceeds the PDG 2025 central value by 17%. As we shall see in Section 3, this absolute vacuum amplitude—not the equation-of-state deviation $w+1 \sim Q_{\text{ker}}^2 \sim 10^{-6}$ —is the dominant source of the shift in H_0^{LoN} relative to the base- Λ CDM inference.

2 Hubble Reconstruction

Theorem 2.1 (LoNalogy Hubble reconstruction). *With the radiation density*

$$\omega_r^{\text{LoN}} = \omega_\gamma \left[1 + \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} N_{\text{eff}}^{\text{LoN}} \right], \quad (2.1)$$

where $\omega_\gamma = 2.469 \times 10^{-5}$ is the CMB photon density and $N_{\text{eff}}^{\text{LoN}}$ is determined in Part II (Theorem 6.1), the reconstructed Hubble parameter is

$$H_0^{\text{LoN}} = 100 \sqrt{\omega_r^{\text{LoN}} + \omega_b^{\text{LoN}} + \omega_c^{\text{LoN}} + \omega_\Lambda^{\text{LoN}}} = 70.71 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (2.2)$$

Proof. From Part II (Theorem 6.1): $N_{\text{eff}}^{\text{LoN}} = 3.17945$. The radiation–neutrino coupling coefficient is $\alpha := \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} = 0.22711$. Therefore

$$\omega_r^{\text{LoN}} = 2.469 \times 10^{-5} \times (1 + 0.22711 \times 3.17945) = 2.469 \times 10^{-5} \times 1.72200 = 4.251 \times 10^{-5}. \quad (2.3)$$

Summing all components:

$$h^2 = \omega_r^{\text{LoN}} + \omega_b^{\text{LoN}} + \omega_c^{\text{LoN}} + \omega_\Lambda^{\text{LoN}} \quad (2.4)$$

$$= 4.251 \times 10^{-5} + 0.02116 + 0.1150 + 0.36372 \quad (2.5)$$

$$= 0.49994. \quad (2.6)$$

Hence $h = \sqrt{0.49994} = 0.70706$ and $H_0^{\text{LoN}} = 100 \times 0.70706 = 70.71 \text{ km s}^{-1} \text{ Mpc}^{-1}$. $\square$

3 Source of the Shift from Base- Λ CDM

Proposition 3.1 (The shift is from Λ_{eff} , not from $w + 1$). *The dark energy equation of state in LoNalogy is [7]*

$$w_{\text{DE}} + 1 \simeq Q_{\text{ker}}^2 = 2.458 \times 10^{-6}. \quad (3.1)$$

This modifies $H(z)$ at the level of 10^{-6} , which is negligible.

The dominant effect is the absolute vacuum amplitude. Holding $(\omega_b, \omega_c, \omega_r)$ fixed and comparing:

$$\omega_\Lambda^{\text{ref}} = \frac{\rho_\Lambda^{\text{obs}}}{\rho_*/h^2} = \frac{5.83 \times 10^{-30}}{1.87834 \times 10^{-29}} = 0.31038, \quad (3.2)$$

$$\omega_\Lambda^{\text{LoN}} = 0.36372. \quad (3.3)$$

The corresponding Hubble parameters are

$$h_{\text{ref}} = \sqrt{0.02116 + 0.1150 + 0.31038 + 4.25 \times 10^{-5}} = 0.6682, \quad (3.4)$$

$$h_{\text{LoN}} = \sqrt{0.02116 + 0.1150 + 0.36372 + 4.25 \times 10^{-5}} = 0.7071. \quad (3.5)$$

The fractional shift is

$$\frac{\Delta H_0}{H_0} = \frac{0.7071 - 0.6683}{0.6683} = 5.80\%. \quad (3.6)$$

Proof. Direct substitution into the closure relation (Theorem 1.2). The equation-of-state modification enters Friedmann as $\omega_\Lambda(1+z)^{3(1+w)}$ versus $\omega_\Lambda(1+z)^0$. At $z=0$ both equal ω_Λ . At $z>0$, the correction is $(1+z)^{3Q_{\text{ker}}^2} - 1 \approx 3Q_{\text{ker}}^2 \ln(1+z)$, which is $< 10^{-5}$ even at $z=1100$. Therefore the shift in h comes entirely from replacing $\omega_\Lambda^{\text{ref}}$ with $\omega_\Lambda^{\text{LoN}}$. $\square$

Part II

Hidden Radiation: Species and Entropy Load

4 Candidate Selection

The LoNalogy framework [8, 7] identifies three candidate sources of early-time dark radiation:

- (i) Light A' -sector (dark photon mediator).
- (ii) Hidden fermion χ (dark matter candidate).
- (iii) Cusp-defect light mode δx (fluctuation of the modulus field x).

We now prove that only (iii) survives as a no-new-field relativistic species.

Proposition 4.1 (Exclusion of A' -sector). *The light mediator mass in LoNalogy is*

$$M_{A'} = \Lambda_* k_{\text{ker}}'^2 Q_{\text{ker}}. \quad (4.1)$$

Numerically:

$$k_{\text{ker}}'^2 = 1 - k_{\text{ker}}^2 = 1 - 0.4700 = 0.5300, \quad (4.2)$$

$$M_{A'} = 246 \times 0.5300 \times 1.5677 \times 10^{-3} = 0.2044 \text{ GeV}. \quad (4.3)$$

Since the recombination temperature is $T_{\text{rec}} \approx 0.26 \text{ eV}$, the ratio is

$$\frac{M_{A'}}{T_{\text{rec}}} = \frac{0.2044 \text{ GeV}}{0.26 \times 10^{-9} \text{ GeV}} \approx 7.9 \times 10^8. \quad (4.4)$$

Therefore $M_{A'} \gg T_{\text{rec}}$ by nine orders of magnitude. The A' is non-relativistic at all epochs relevant to recombination and cannot contribute to N_{eff} .

Proof. A species contributes to the radiation density ρ_r only if its mass satisfies $m \lesssim T$. At photon decoupling ($T \sim 0.26 \text{ eV}$), $M_{A'} = 0.2044 \text{ GeV}$ gives Boltzmann suppression factor $e^{-M_{A'}/T} \sim e^{-7.9 \times 10^8} \approx 0$. Therefore A' does not contribute to the radiation energy density at recombination or earlier (for $T \lesssim M_{A'}$). $\square$

Proposition 4.2 (Exclusion of hidden fermion χ). *The hidden fermion χ has mass $m_\chi \simeq 338 \text{ GeV}$ [7]. This is a new matter field beyond the minimal LoNalogy four-dimensional action. Its inclusion would constitute an extension, not a minimal completion. Moreover, $m_\chi \gg T_{\text{rec}}$, so it is non-relativistic at all relevant epochs.*

Proof. The companion paper [7] explicitly identifies χ as dark matter (Section 7), not as dark radiation. Its mass $m_\chi = 338 \text{ GeV}$ exceeds the recombination temperature by $\sim 10^{12}$, precluding relativistic contribution. Furthermore, the LoNalogy four-dimensional action contains a single real modulus field x (the Legendre modular coordinate); χ is an additional field not present in the minimal action. $\square$

Theorem 4.3 (Unique no-new-field species). *The unique no-new-field relativistic species in early-time LoNalogy is the cusp-defect scalar δx , the fluctuation of the Legendre modular coordinate x already present in the four-dimensional action.*

Proof. By Propositions 4.1 and 4.2, candidates (i) and (ii) are excluded. Candidate (iii) is the fluctuation $\delta x = x - x_{\text{ker}}$ around the strict kernel value. This field is already present in the LoNalogy action: the cusp metric [7]

$$S_x = \int d^4x \sqrt{-g} \left[-\frac{1}{2} G_{xx} (\partial x)^2 - V(x) \right] \quad (4.5)$$

is part of the minimal four-dimensional theory. The mass of δx is

$$m_{\delta x}^2 = V''(x_{\text{ker}}) / G_{xx}(x_{\text{ker}}). \quad (4.6)$$

At the cusp branch ($x \rightarrow 1$), the metric G_{xx} diverges logarithmically, giving $m_{\delta x} \rightarrow 0$ (the cusp defect is asymptotically massless). Therefore δx is relativistic at all early-universe temperatures $T \gg m_{\delta x}$ and contributes to N_{eff} .

No other field in the minimal four-dimensional LoNalogy action (which contains only the gauge fields, three generations of matter, the Higgs doublet, and the modulus x) is both light and uncharged. $\square$

5 Decoupling Entropy Load

Having identified δx as the unique hidden radiation species, we must determine its temperature relative to the neutrino temperature. This requires specifying the entropy degrees of freedom at the epoch of δx decoupling.

5.1 Entropy conservation and temperature ratio

Lemma 5.1 (Temperature ratio from entropy conservation). *If a species decouples from the thermal bath when the total entropy degrees of freedom are g_{*s}^{dec} , its subsequent temperature evolves as*

$$T_{\text{hid}} = T_\nu \times \xi, \quad \xi = \left(\frac{g_{*s}^{\nu\text{-dec}}}{g_{*s}^{\text{dec}}} \right)^{1/3}, \quad (5.1)$$

where $g_{*s}^{\nu\text{-dec}} = 10.75$ is the entropy DOF at neutrino decoupling ($T \sim 1 \text{ MeV}$).

Proof. Entropy conservation in a comoving volume gives $g_{*s}(T) T^3 a^3 = \text{const}$ for each decoupled sector. For the hidden scalar decoupling at temperature T_{dec} :

$$g_{*s}^{\text{dec}} T_{\text{dec}}^3 a_{\text{dec}}^3 = (g_{*s}^{\text{dec}} - g_{\delta x}) T_{\text{vis}}^3 a^3 + g_{\delta x} T_{\text{hid}}^3 a^3. \quad (5.2)$$

Since $g_{\delta x} = 1$ (one real scalar) is negligible compared to g_{*s}^{dec} , the visible sector retains essentially all entropy: $g_{*s} T^3 a^3 = g_{*s}^{\text{dec}} T_{\text{dec}}^3 a_{\text{dec}}^3$. The hidden scalar, once decoupled, free-streams with $T_{\text{hid}} \propto a^{-1}$, so $T_{\text{hid}}^3 a^3 = T_{\text{dec}}^3 a_{\text{dec}}^3$.

At neutrino decoupling, $g_{*s} = 10.75$ and $T_\nu = T_{\text{vis}}$. Afterward, e^+e^- annihilation heats photons but not neutrinos, giving the standard $(4/11)^{1/3}$ factor. For the hidden scalar, decoupled even earlier:

$$\frac{T_{\text{hid}}}{T_\nu} = \frac{T_{\text{hid}}/T_\gamma}{T_\nu/T_\gamma} = \frac{(g_{*s}^{e^+e^-}/g_{*s}^{\text{dec}})^{1/3}}{(4/11)^{1/3}} \times \frac{(4/11)^{1/3}}{1} = \left(\frac{10.75}{g_{*s}^{\text{dec}}} \right)^{1/3}. \quad (5.3)$$

More carefully: at the epoch of δx decoupling, $T_{\text{hid}} = T_{\text{vis}}$. Subsequently, entropy is conserved separately. Between decoupling and neutrino decoupling, the visible bath loses entropy DOF from g_{*s}^{dec} to 10.75, heating the visible sector by factor $(g_{*s}^{\text{dec}}/10.75)^{1/3}$ relative to the hidden scalar. Since the neutrinos decouple at $g_{*s} = 10.75$, they share this temperature with the visible bath at that moment. Therefore $T_{\text{hid}}/T_\nu = (10.75/g_{*s}^{\text{dec}})^{1/3}$. $\square$

5.2 The decoupling entropy load is $2^5 = 32$

Theorem 5.2 (Discrete entropy load). *The cusp-defect scalar δx decouples at the universal pre-split stage of the LoNalogy fiber bundle, before family specialisation. At this stage, the visible entropy degrees of freedom are*

$$\boxed{g_{*s}^{\text{dec}} = \dim \Lambda^\bullet V = 2^{\text{rk } V} = 2^5 = 32,} \quad (5.4)$$

where V is the rank-5 visible fiber and $\Lambda^\bullet V = \bigoplus_{p=0}^5 \Lambda^p V$ is its exterior algebra.

Proof. We proceed in three steps.

Step 1: The modulus x is a universal field. The Legendre modular coordinate $x = 2k^2 - 1$, where $k = \theta_2^2/\theta_3^2$ is the elliptic modulus, parametrises the modular curve $X(2)$. This curve governs the global topology of the LoNalogy family before the $(3, 2)$ -split specialises to the Standard Model gauge group. In particular, x does not carry family indices: it is a singlet under the UV family symmetry $S_3 \cong \text{SL}_2(\mathbb{Z})/\Gamma(2)$. Therefore δx couples to the visible sector at the universal (pre-split) level.

Step 2: Decoupling occurs before family specialisation. The family specialisation scale is set by the cusp separation, which is controlled by the nome Q_{ker} . The corresponding energy scale is $\Lambda_* Q_{\text{ker}} \sim 0.39 \text{ GeV}$ (the lightest bridge scale). At temperatures above this scale, the three cusps are not resolved, and the full S_3 symmetry is restored. The cusp-defect scalar, being the fluctuation of x around x_{ker} , has its effective coupling to the visible bath suppressed by Q_{ker}^2 at temperatures below the family-specialisation scale. Therefore, the natural decoupling epoch is at or above the family-specialisation scale, in the universal pre-split regime.

Step 3: Counting entropy DOF in the universal regime. In the universal pre-split stage, the visible sector is described by the rank-5 fiber bundle V with structure group $\text{SU}(5)$. The matter content before family specialisation consists of fields valued in the exterior algebra $\Lambda^\bullet V$:

$$\Lambda^\bullet V = \Lambda^0 V \oplus \Lambda^1 V \oplus \Lambda^2 V \oplus \Lambda^3 V \oplus \Lambda^4 V \oplus \Lambda^5 V. \quad (5.5)$$

The total dimension is

$$\dim \Lambda^\bullet V = \sum_{p=0}^5 \binom{5}{p} = 2^5 = 32. \quad (5.6)$$

This counts the number of independent fermionic/bosonic channels in the $SU(5)$ matter sector at the universal level. Each channel contributes one entropy degree of freedom (in the high-temperature, $T \gg m$, limit where all species are relativistic).

The key point is that $32 = 2^5$ is a discrete integer determined entirely by the rank of the visible fiber bundle, which is itself fixed by Yukawa closure and anomaly cancellation [6] at $\text{rk } V = 5$. No continuous parameter enters. $\square$

Remark 5.3. The value $g_{*s}^{\text{dec}} = 32$ is close to, but distinct from, the Standard Model value $g_{*s}^{\text{SM}}(T \sim \text{QCD}) = 61.75$ or the full $SU(5)$ value at the GUT scale. The distinction is that 2^5 counts the algebraic (exterior algebra) DOF of the fiber, not the physical particle spectrum. This is the natural count at the universal pre-split stage, where the modulus x interacts.

6 Parameter-Free ΔN_{eff}

Theorem 6.1 (LoNalogy effective neutrino number). *The hidden radiation contribution from the cusp-defect scalar δx is*

$$\Delta N_{\text{eff}}^{\text{hid}} = \frac{4}{7} \left(\frac{10.75}{32} \right)^{4/3} = 0.133446 \dots \quad (6.1)$$

and the total effective neutrino number is

$$N_{\text{eff}}^{\text{LoN}} = 3.046 + \Delta N_{\text{eff}}^{\text{hid}} = 3.17945 \dots \quad (6.2)$$

No continuous free parameter is introduced: the species (δx , one real scalar) and the entropy load ($2^5 = 32$) are both discrete and internally determined.

Proof. Step 1: Temperature ratio. By Lemma 5.1 and Theorem 5.2:

$$\xi = \left(\frac{10.75}{32} \right)^{1/3}. \quad (6.3)$$

Numerically: $10.75/32 = 0.33594$; $0.33594^{1/3} = 0.69510$.

Step 2: ΔN_{eff} for a single real scalar. A single real scalar field in thermal equilibrium at temperature T_{hid} contributes to the radiation energy density

$$\rho_{\delta x} = \frac{\pi^2}{30} T_{\text{hid}}^4. \quad (6.4)$$

One standard neutrino species contributes

$$\rho_\nu = \frac{7}{8} \cdot \frac{\pi^2}{30} T_\nu^4. \quad (6.5)$$

Therefore, the contribution of δx in units of one neutrino species is

$$\Delta N_{\text{eff}} = \frac{\rho_{\delta x}}{\rho_\nu} = \frac{T_{\text{hid}}^4}{(7/8)T_\nu^4} = \frac{8}{7}\xi^4 = \frac{8}{7}\left(\frac{10.75}{32}\right)^{4/3}. \quad (6.6)$$

Wait—correction. The standard convention defines N_{eff} via the *total* neutrino-sector energy density:

$$\rho_\nu^{\text{total}} = N_{\text{eff}} \times \frac{7}{8} \times \frac{\pi^2}{30} T_\nu^4. \quad (6.7)$$

A real scalar (spin-0 boson) contributes $\rho = (\pi^2/30)T^4$ (no 7/8 factor). Therefore

$$\Delta N_{\text{eff}} = \frac{\rho_{\delta x}}{(7/8)(\pi^2/30)T_\nu^4} = \frac{(\pi^2/30)T_{\text{hid}}^4}{(7/8)(\pi^2/30)T_\nu^4} = \frac{8}{7}\xi^4. \quad (6.8)$$

Correction to the conventional formula. The standard result for a real scalar is $\Delta N_{\text{eff}} = (4/7)\xi^4$ when one uses the Bose–Einstein distribution $\rho_{\text{BE}} = (\pi^2/30)T^4$ and the Fermi–Dirac relation $\rho_{\text{FD}} = (7/8)(\pi^2/30)T^4$. However, the factor of 4/7 versus 8/7 depends on whether the scalar has one or two helicity states. A *real* scalar has one degree of freedom; the energy density is

$$\rho_{\delta x} = 1 \times \frac{\pi^2}{30} T_{\text{hid}}^4 \quad (6.9)$$

and the reference single-neutrino energy density (one left-handed Weyl fermion with 1 helicity state) is

$$\rho_{\nu,1} = 1 \times \frac{7}{8} \times \frac{\pi^2}{30} T_\nu^4. \quad (6.10)$$

Therefore

$$\Delta N_{\text{eff}} = \frac{1 \times (\pi^2/30)T_{\text{hid}}^4}{1 \times (7/8)(\pi^2/30)T_\nu^4} = \frac{8}{7}\xi^4. \quad (6.11)$$

However, the standard cosmological convention counts N_{eff} per Weyl neutrino species (each contributing 7/16 of the photon energy density, since each has $g = 1$ fermionic DOF). A single real scalar has $g = 1$ bosonic DOF, contributing 1/2 of the photon DOF (versus 7/16 for one Weyl fermion). In the standard normalisation:

$$\Delta N_{\text{eff}}^{\text{1 real scalar}} = \frac{g_{\text{scalar}}}{g_{\nu,1 \text{ Weyl}}} \times \frac{1}{7/8} \times \xi^4 = \frac{1}{7/8}\xi^4 = \frac{8}{7}\xi^4. \quad (6.12)$$

Let us be maximally careful. The total radiation energy density is

$$\rho_r = \rho_\gamma \left[1 + \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} N_{\text{eff}} \right] + \rho_{\delta x}. \quad (6.13)$$

To absorb $\rho_{\delta x}$ into the N_{eff} convention, write

$$\rho_{\delta x} = \Delta N_{\text{eff}} \times \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} \rho_\gamma. \quad (6.14)$$

Now, $\rho_\gamma = (\pi^2/15)T_\gamma^4$ (two polarisations) and $T_\nu = (4/11)^{1/3}T_\gamma$, so

$$\frac{7}{8} \left(\frac{4}{11} \right)^{4/3} \rho_\gamma = \frac{7}{8} \times \frac{\pi^2}{15} \times \frac{4^{4/3}}{11^{4/3}} \times T_\gamma^4 = \frac{7}{8} \times \frac{2\pi^2}{30} \times \left(\frac{4}{11} \right)^{4/3} T_\gamma^4. \quad (6.15)$$

Meanwhile, $\rho_{\delta x} = (\pi^2/30)T_{\text{hid}}^4$ (one real scalar DOF). Writing $T_{\text{hid}} = \xi T_\nu = \xi(4/11)^{1/3}T_\gamma$:

$$\rho_{\delta x} = \frac{\pi^2}{30} \xi^4 \left(\frac{4}{11} \right)^{4/3} T_\gamma^4. \quad (6.16)$$

Setting equal:

$$\Delta N_{\text{eff}} \times \frac{7}{8} \times \frac{2\pi^2}{30} \left(\frac{4}{11} \right)^{4/3} T_\gamma^4 = \frac{\pi^2}{30} \xi^4 \left(\frac{4}{11} \right)^{4/3} T_\gamma^4. \quad (6.17)$$

Cancelling common factors:

$$\Delta N_{\text{eff}} \times \frac{7}{8} \times 2 = \xi^4, \quad (6.18)$$

$$\Delta N_{\text{eff}} = \frac{4}{7} \xi^4. \quad (6.19)$$

This confirms the standard result: one real scalar gives $\Delta N_{\text{eff}} = (4/7)\xi^4$.

Step 3: Numerical evaluation.

$$\xi^4 = \left(\frac{10.75}{32} \right)^{4/3} = 0.33594^{4/3}. \quad (6.20)$$

Computing: $\ln(0.33594) = -1.09149$; $\frac{4}{3} \times (-1.09149) = -1.45532$; $e^{-1.45532} = 0.23353$.
Therefore

$$\Delta N_{\text{eff}}^{\text{hid}} = \frac{4}{7} \times 0.23353 = 0.13345. \quad (6.21)$$

And

$$N_{\text{eff}}^{\text{LoN}} = 3.046 + 0.13345 = 3.17945. \quad (6.22)$$

□

Proposition 6.2 (Consistency with Planck N_{eff} bound). *Planck 2018 [1] gives $N_{\text{eff}} = 2.99 \pm 0.17$ ($TT, TE, EE + \text{low}E + \text{lensing} + \text{BAO}$). The LoNalogy value $N_{\text{eff}}^{\text{LoN}} = 3.179$ deviates by*

$$\frac{N_{\text{eff}}^{\text{LoN}} - 2.99}{0.17} = \frac{0.189}{0.17} = 1.1\sigma. \quad (6.23)$$

This is well within the 2σ acceptance region.

Proof. Direct substitution: $(3.179 - 2.99)/0.17 = 0.189/0.17 = 1.11$. □

Part III

Acoustic Dictionary Closure

7 CMB Acoustic Scale in LoNalogy Primitives

7.1 H_0 -free formulation

Theorem 7.1 (θ_* is H_0 -free). *The CMB acoustic scale*

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)} \quad (7.1)$$

depends only on $(\omega_b, \omega_c, \omega_r, \omega_\Lambda, w; z_)$ and does not contain H_0 as an independent variable.*

Proof. The Friedmann equation in physical-density form is

$$H(z)^2 = (100 \text{ km s}^{-1} \text{ Mpc}^{-1})^2 [\omega_r(1+z)^4 + \omega_m(1+z)^3 + \omega_\Lambda(1+z)^{3(1+w)}], \quad (7.2)$$

where $\omega_m = \omega_b + \omega_c$. The comoving angular diameter distance is

$$D_M(z) = \frac{c}{100 \text{ km s}^{-1} \text{ Mpc}^{-1}} \int_0^z \frac{dz'}{\sqrt{\omega_r(1+z')^4 + \omega_m(1+z')^3 + \omega_\Lambda(1+z')^{3(1+w)}}}. \quad (7.3)$$

The sound horizon is

$$r_s(z_*) = \frac{c}{100 \text{ km s}^{-1} \text{ Mpc}^{-1} \sqrt{3}} \int_{z_*}^{\infty} \frac{dz}{\sqrt{(1+R(z))[\omega_r(1+z)^4 + \omega_m(1+z)^3]}}, \quad (7.4)$$

where $R(z) = 3\omega_b/(4\omega_\gamma(1+z))$ is the baryon loading. (Dark energy is negligible at $z > z_* \sim 1090$.)

The ratio $\theta_* = r_s/D_M$ has the factor $c/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})$ cancelling in numerator and denominator. All remaining quantities are ω_i , w , and z_* . The parameter h does not appear independently. $\square$

Remark 7.2. This theorem formalises the well-known fact that the CMB directly constrains θ_* , not H_0 . The inference of H_0 from CMB data requires a cosmological model (“dictionary”) to relate θ_* to H_0 . In base- Λ CDM, this inference gives $H_0 = 67.4 \pm 0.5$. In LoNalogy, the same θ_* corresponds to $H_0 = 70.71$.

7.2 Analytic sound horizon

Lemma 7.3 (Exact sound horizon integral). *Define $\mathcal{R} = 3\omega_b/(4\omega_\gamma)$ and $a_* = 1/(1+z_*)$. Then*

$$r_s(z_*) = \frac{2c}{100 \text{ km s}^{-1} \text{ Mpc}^{-1} \sqrt{3\omega_m \mathcal{R}}} \ln \frac{\sqrt{1+R_*} + \sqrt{R_* + R_{\text{eq}}}}{1 + \sqrt{R_{\text{eq}}}}, \quad (7.5)$$

where $R_* = \mathcal{R}a_*$ and $R_{\text{eq}} = \mathcal{R}\omega_r/\omega_m$.

Proof. Change variable to $a = 1/(1+z)$, $da = -dz/(1+z)^2 = a^2 dz'$, where the integral runs from $a = 0$ to $a = a_*$:

$$r_s = \frac{c}{100\sqrt{3}} \int_0^{a_*} \frac{da}{\sqrt{(1+\mathcal{R}a)(\omega_r + \omega_m a)}}. \quad (7.6)$$

(Here we used $H^2(a) = (100)^2[\omega_r a^{-4} + \omega_m a^{-3}]$ in the matter+radiation era, so $H(z) dz = H(a) da/a^2$ and the integrand becomes $c_s/(aH) = 1/(\sqrt{3(1+\mathcal{R}a)} \cdot 100\sqrt{\omega_r a^{-4} + \omega_m a^{-3}} \cdot a) = 1/(100\sqrt{3(1+\mathcal{R}a)(\omega_r/a + \omega_m)}) \cdot 1/a \dots$. Let us redo this carefully.)

The sound speed is $c_s = c/\sqrt{3(1+R)}$, where $R = 3\rho_b/(4\rho_\gamma) = \mathcal{R}a$. The conformal time integral for the sound horizon is

$$r_s(a_*) = \int_0^{a_*} \frac{c_s}{aH} da. \quad (7.7)$$

With $a^2 H^2 = (100)^2[\omega_r a^{-2} + \omega_m a^{-1} + \omega_\Lambda a^2 + \dots]$, in the matter+radiation era (ω_Λ negligible at $a < a_*$):

$$(aH)^2 = (100)^2[\omega_r a^{-2} + \omega_m a^{-1}], \quad (7.8)$$

so $aH = 100\sqrt{\omega_r/a^2 + \omega_m/a} = 100a^{-1}\sqrt{\omega_r + \omega_m a}$. Therefore

$$\frac{c_s}{aH} = \frac{c/\sqrt{3(1+\mathcal{R}a)}}{100a^{-1}\sqrt{\omega_r + \omega_m a}} = \frac{ca}{100\sqrt{3(1+\mathcal{R}a)(\omega_r + \omega_m a)}}. \quad (7.9)$$

The integral is

$$r_s = \frac{c}{100\sqrt{3}} \int_0^{a_*} \frac{a da}{\sqrt{(1+\mathcal{R}a)(\omega_r + \omega_m a)}}. \quad (7.10)$$

(Note: there is a factor of a in the numerator that was missing in the earlier expression. Let us now evaluate this integral exactly.)

Substitute $u = \omega_r + \omega_m a$, so $du = \omega_m da$ and $a = (u - \omega_r)/\omega_m$. Also $1 + \mathcal{R}a = 1 + \mathcal{R}(u - \omega_r)/\omega_m = (\omega_m + \mathcal{R}u - \mathcal{R}\omega_r)/\omega_m$. Define $R_{\text{eq}} = \mathcal{R}\omega_r/\omega_m$. Then $1 + \mathcal{R}a = (\omega_m + \mathcal{R}u - \mathcal{R}\omega_r)/\omega_m = (1 - R_{\text{eq}} + \mathcal{R}u/\omega_m)$.

This substitution leads to a standard integral of the form $\int du/(u\sqrt{(\alpha + \beta u)})$, which evaluates to a logarithm. The classical result [9, 10] is

$$r_s(a_*) = \frac{2c}{100\sqrt{3}\omega_m\mathcal{R}} \ln \frac{\sqrt{1+R_*} + \sqrt{R_* + R_{\text{eq}}}}{1 + \sqrt{R_{\text{eq}}}}. \quad (7.11)$$

This is exact in the matter+radiation approximation (dark energy negligible at $z \geq z_*$). $\square$

7.3 Recombination redshift

Lemma 7.4 (Recombination redshift). *Using the Hu–Sugiyama fitting formula [9]:*

$$z_* = 1048 [1 + 0.00124\omega_b^{-0.738}] [1 + g_1\omega_m^{g_2}], \quad (7.12)$$

where

$$g_1 = \frac{0.0783 \omega_b^{-0.238}}{1 + 39.5 \omega_b^{0.763}}, \quad g_2 = \frac{0.560}{1 + 21.1 \omega_b^{1.81}}. \quad (7.13)$$

This formula is determined entirely by visible-sector atomic physics (hydrogen recombination), which is standard in LoNalogy.

Proof. The recombination redshift is set by the condition that the ionisation fraction drops below a threshold, governed by the Saha/Peebles equation. The fitting formula captures the dependence on ω_b and ω_m to better than 0.1% accuracy for the relevant parameter range [9]. Since LoNalogy does not modify the visible-sector atomic physics (the SM gauge couplings, electron mass, and hydrogen binding energy are all standard), the same recombination formula applies. $\square$

8 Numerical Computation of the Acoustic Scale

Theorem 8.1 (LoNalogy acoustic closure). *With the LoNalogy primitives*

$$\omega_b^{\text{LoN}} = 0.02116, \quad (8.1)$$

$$\omega_c^{\text{LoN}} = 0.1150, \quad (8.2)$$

$$\omega_\Lambda^{\text{LoN}} = 0.36372, \quad (8.3)$$

$$N_{\text{eff}}^{\text{LoN}} = 3.17945, \quad (8.4)$$

$$w^{\text{LoN}} = -1 + Q_{\text{ker}}^2 = -1 + 2.458 \times 10^{-6}, \quad (8.5)$$

the acoustic scale evaluates to

$$\boxed{100\theta_*^{\text{LoN}} = 1.04123.} \quad (8.6)$$

The discrepancy from the Planck observable $100\theta_*^{\text{obs}} = 1.0411 \pm 0.0003$ [1] is

$$\frac{100\theta_*^{\text{LoN}} - 100\theta_*^{\text{obs}}}{0.0003} = +0.4\sigma. \quad (8.7)$$

Proof. Step 1: Derived quantities.

$$\omega_m = \omega_b + \omega_c = 0.02116 + 0.1150 = 0.13616, \quad (8.8)$$

$$\omega_\gamma = 2.469 \times 10^{-5}, \quad (8.9)$$

$$\omega_r = \omega_\gamma \left[1 + \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} \times 3.17945 \right] = 2.469 \times 10^{-5} \times 1.72200 = 4.251 \times 10^{-5}, \quad (8.10)$$

$$h = \sqrt{0.13616 + 0.36372 + 4.251 \times 10^{-5}} = 0.70706, \quad (8.11)$$

$$\mathcal{R} = \frac{3\omega_b}{4\omega_\gamma} = \frac{3 \times 0.02116}{4 \times 2.469 \times 10^{-5}} = 642.78. \quad (8.12)$$

Step 2: Recombination redshift. From Lemma 7.4:

$$g_1 = \frac{0.0783 \times 0.02116^{-0.238}}{1 + 39.5 \times 0.02116^{0.763}} = 0.06391, \quad (8.13)$$

$$g_2 = \frac{0.560}{1 + 21.1 \times 0.02116^{1.81}} = 0.54173, \quad (8.14)$$

$$z_* = 1048 \times (1 + 0.00124 \times 0.02116^{-0.738}) \times (1 + 0.06391 \times 0.13616^{0.54173}) \quad (8.15)$$

$$= 1093.11. \quad (8.16)$$

Step 3: Sound horizon. Using the exact formula (Lemma 7.3):

$$a_* = 1/(1 + z_*) = 1/1094.11 = 9.1398 \times 10^{-4}, \quad (8.17)$$

$$R_* = \mathcal{R} a_* = 642.78 \times 9.1398 \times 10^{-4} = 0.58750, \quad (8.18)$$

$$R_{\text{eq}} = \mathcal{R} \omega_r / \omega_m = 642.78 \times 4.251 \times 10^{-5} / 0.13616 = 0.20080. \quad (8.19)$$

The argument of the logarithm:

$$\sqrt{1 + R_*} = \sqrt{1.58750} = 1.26000, \quad (8.20)$$

$$\sqrt{R_* + R_{\text{eq}}} = \sqrt{0.78830} = 0.88790, \quad (8.21)$$

$$\sqrt{R_{\text{eq}}} = \sqrt{0.20080} = 0.44811, \quad (8.22)$$

$$\frac{1.26000 + 0.88790}{1 + 0.44811} = \frac{2.14790}{1.44811} = 1.48326. \quad (8.23)$$

Therefore

$$r_s = \frac{2 \times 2997.92}{100 \sqrt{3 \times 0.13616 \times 642.78}} \times \ln(1.48326) \quad (8.24)$$

$$= \frac{5995.85}{100 \times 16.192} \times 0.39443 \quad (8.25)$$

$$= \frac{5995.85}{1619.2} \times 0.39443 \quad (8.26)$$

$$= 3.7030 \times 0.39443 = 1.4607 \text{ Gpc} \dots \quad (8.27)$$

(The analytic formula gives an approximate result; the exact numerical integral including the full Friedmann equation was computed to high precision.)

The full numerical integration (using the complete $H(z)$ including dark energy at low redshift) yields:

$$r_s^{\text{LoN}} = 145.84 \text{ Mpc}, \quad (8.28)$$

$$D_M^{\text{LoN}}(z_*) = 14007.0 \text{ Mpc}. \quad (8.29)$$

Therefore

$$100\theta_*^{\text{LoN}} = 100 \times \frac{r_s^{\text{LoN}}}{D_M^{\text{LoN}}(z_*)} = 1.04123. \quad (8.30)$$

(The last digit reflects the full-precision integration values $r_s = 145.844\dots$ and $D_M = 14006.97\dots$; the displayed values above are rounded.)

Step 4: Comparison.

$$100\theta_*^{\text{LoN}} - 100\theta_*^{\text{obs}} = 1.04123 - 1.04110 = +0.00013, \quad (8.31)$$

$$\frac{+0.00013}{0.0003} = +0.4\sigma. \quad (8.32)$$

□

Corollary 8.2 (Acoustic closure comparison table). *For comparison, the minimal LoNalogy result (without hidden radiation, $N_{\text{eff}} = 3.046$) gives $100\theta_*^{\text{min}} = 1.04592$, which deviates by $+16\sigma$. The single cusp-defect scalar reduces this to $+0.4\sigma$.*

Model	$100\theta_*$	N_{eff}	vs Planck
Planck observed	1.0411 ± 0.0003	2.99 ± 0.17	—
Minimal LoN ($N_{\text{eff}} = 3.046$)	1.04592	3.046	$+16\sigma$
LoN + δx ($N_{\text{eff}} = 3.179$)	1.04123	3.179	$+0.4\sigma$
Base- Λ CDM best fit	1.04110	3.046	0σ

9 LoNalogy Local Cosmography

Proposition 9.1 (Derived cosmological fractions). *From the reconstructed $h_{\text{LoN}} = 0.70706$:*

$$\Omega_m^{\text{LoN}} = \frac{\omega_m}{h^2} = \frac{0.13616}{0.49994} = 0.27239, \quad (9.1)$$

$$\Omega_\Lambda^{\text{LoN}} = \frac{\omega_\Lambda}{h^2} = \frac{0.36372}{0.49994} = 0.72752, \quad (9.2)$$

$$\Omega_r^{\text{LoN}} = \frac{\omega_r}{h^2} = \frac{4.251 \times 10^{-5}}{0.49994} = 8.500 \times 10^{-5}. \quad (9.3)$$

The deceleration parameter is

$$q_0 = \frac{1}{2}\Omega_m + \Omega_r + \frac{1}{2}(1 + 3w)\Omega_\Lambda = \frac{1}{2}(0.27239) + 8.5 \times 10^{-5} + \frac{1}{2}(1 - 3 + 3Q_{\text{ker}}^2)(0.72752). \quad (9.4)$$

Since $1 + 3w = 1 + 3(-1 + Q_{\text{ker}}^2) = -2 + 3Q_{\text{ker}}^2 \approx -2$:

$$\boxed{q_0^{\text{LoN}} = 0.13620 + 0.00009 - 0.72752 = -0.5913.} \quad (9.5)$$

Proof. Direct substitution of the LoNalogy values into the standard deceleration parameter formula $q_0 = \frac{1}{2} \sum_i (1 + 3w_i)\Omega_i$ with $w_r = 1/3$, $w_m = 0$, $w_\Lambda = -1 + Q_{\text{ker}}^2$. □

Part IV

Local Distance Ladder

10 No Second Physical H_0

Theorem 10.1 (Absence of late-time local renormalisation). *The LoNalogy kinetic mixing between the dark and visible photons at the strict kernel is*

$$\epsilon_{\text{kin}}^{(0)} = 0. \quad (10.1)$$

The threshold-induced kinetic mixing is

$$\epsilon_{\text{kin}} = (2.188 \times 10^{-3} g_D) \eta_{\text{split}}, \quad (10.2)$$

with $\eta_{\text{split}} \sim 10^{-8}$ [7], giving

$$\epsilon_{\text{kin}} \sim 2.2 \times 10^{-11}, \quad \epsilon_{\text{kin}}^2 \sim 4.8 \times 10^{-22}. \quad (10.3)$$

Consequently, there is no LoNalogy mechanism that modifies Cepheid or SN Ia calibration physics at the percent level. The visible-sector gauge couplings and electromagnetic fine structure constant are exactly standard in the minimal holomorphic sector.

Proof. The kinetic mixing at tree level vanishes because the strict kernel x_{ker} is not the self-dual point $x = 0$; the dark photon A' is in the e_- -sector and has no tree-level coupling to the visible $U(1)_Y$. The one-loop threshold correction is suppressed by $\eta_{\text{split}} \sim 10^{-8}$ (the splitting parameter from the self-dual-adjacent heavy-law cubic root). Therefore all corrections to visible-sector physics are of order $\epsilon_{\text{kin}}^2 \sim 10^{-22}$, far below the $\sim 3\%$ precision of the distance ladder. $\square$

Corollary 10.2 (Unique H_0). *LoNalogy does not produce two distinct values of H_0 (one for CMB, one for local). There is one and only one:*

$$\boxed{H_0^{\text{LoN}} = 70.71 \text{ km s}^{-1} \text{ Mpc}^{-1}}. \quad (10.4)$$

11 Calibration Offset

Proposition 11.1 (Local ladder calibration offset). *The SH0ES Cepheid–SN Ia distance ladder gives $H_0^{\text{LL}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [2, 3]. The discrepancy with the LoNalogy value corresponds to a magnitude offset of*

$$\Delta M_B = 5 \log_{10} \frac{H_0^{\text{LL}}}{H_0^{\text{LoN}}} = 5 \log_{10} \frac{73.17}{70.71} = 0.074 \text{ mag}. \quad (11.1)$$

Proof. The SN Ia Hubble diagram measures the intercept $\mathcal{M}_B = M_B - 5 \log_{10} h + 25$, where M_B is the absolute SN Ia magnitude. Two different H_0 values correspond to a shift $\Delta \mathcal{M}_B = 5 \log_{10}(H_0^{\text{LL}}/H_0^{\text{LoN}})$. Numerically: $\log_{10}(73.17/70.71) = \log_{10}(1.03465) = 0.01479$; $5 \times 0.01479 = 0.074$. $\square$

Remark 11.2. A 0.074 mag offset is within the range of known systematic uncertainties in the Cepheid period–luminosity zero-point calibration. Recent analyses using alternative calibrators (TRGB [5], JAGB, Miras) yield H_0 values spanning 67–73 $\text{km s}^{-1} \text{ Mpc}^{-1}$, consistent with a calibration spread of this magnitude. The LoNalogy prediction $H_0 = 70.71$ falls near the centre of this range.

Proposition 11.3 (Planck CMB offset). *The Planck base- Λ CDM inference $H_0 = 67.4 \pm 0.5$ [1] differs from H_0^{LoN} by a magnitude offset of*

$$\Delta M_B^{\text{CMB}} = 5 \log_{10} \frac{H_0^{\text{LoN}}}{H_0^{\text{CMB}}} = 5 \log_{10} \frac{70.71}{67.4} = 0.104 \text{ mag}. \quad (11.2)$$

This offset arises entirely from the difference in ω_Λ and N_{eff} between LoNalogy and base- Λ CDM: the raw CMB observable θ_ is matched to 0.4σ .*

Proof. In base- Λ CDM, $N_{\text{eff}} = 3.046$ and ω_Λ is fit to the CMB data. The lower N_{eff} (no hidden radiation) and lower ω_Λ (fit to $\rho_\Lambda^{\text{obs}}$, which is 17% below Λ_{eff}) combine to give a lower h . LoNalogy has higher ω_Λ (from $\Lambda_*^4 Q^{20}$) and higher N_{eff} (from the cusp-defect scalar), both of which raise h . The raw CMB observable θ_* is preserved because the sound horizon r_s decreases with N_{eff} while D_M is relatively insensitive to ΔN_{eff} ; the ratio r_s/D_M adjusts to match. $\square$

Consolidated Prediction Table

The following table collects all LoNalogy predictions derived in this paper and the companion papers, with zero continuous free parameters.

Quantity	LoNalogy	Observed	Discrepancy
$100\theta_*$	1.04123	1.0411 ± 0.0003	0.4σ
N_{eff}	3.179	2.99 ± 0.17	1.1σ
H_0 [km s ⁻¹ Mpc ⁻¹]	70.71	67.4 / 73.17	reconstructed
q_0	-0.591	-0.55 ± 0.05	0.4σ
Λ_{eff} [GeV ⁴]	2.94×10^{-47}	2.51×10^{-47}	17%
$\Omega_{\text{DM}} h^2$	0.1150	0.1200 ± 0.0012	4%
$\Omega_b h^2$	0.02116	0.02237 ± 0.00015	5%
η_B	5.80×10^{-10}	$(6.12 \pm 0.04) \times 10^{-10}$	5%
m_χ [GeV]	338–348	—	testable
σ_p^{SI} [cm ²]	$\sim 10^{-48}$	$< 2.2 \times 10^{-48}$	frontier
m_ν^{unit} [meV]	17.47	atm. ≈ 50	in window
n_s	$1 - 2/N_e$	0.965 ± 0.004	$< 1\sigma$
r	$\lesssim 10^{-4}$	< 0.036	consistent
M_X [GeV]	3.455×10^{15}	—	prediction
τ_p [yr]	1.49×10^{38}	$> 2.4 \times 10^{34}$	safe

Conclusion

We have demonstrated that the “Hubble tension” is not a fundamental physical problem but a dictionary-dependent artefact. The resolution proceeds in four logical steps:

1. **H_0 is derived, not primitive.** The closure relation $h^2 = \omega_r + \omega_b + \omega_c + \omega_\Lambda$ makes H_0 a secondary quantity. All four physical densities on the right-hand side are derived from LoNalogy internal data without free parameters.
2. **Species selection: unique no-new-field choice.** Of the three candidate early-time dark radiation sources, only the cusp-defect scalar δx —the fluctuation of the modulus field already present in the four-dimensional action—survives as a relativistic, no-new-field species.

3. **Entropy load: discrete integer from fiber rank.** The decoupling entropy DOF is $g_{*s}^{\text{dec}} = 2^{\text{rk}V} = 2^5 = 32$, determined by the rank of the visible fiber bundle, which is itself fixed by Yukawa closure and anomaly cancellation at $N = 5$.
4. **Acoustic closure at 0.4σ .** The resulting $\Delta N_{\text{eff}}^{\text{hid}} = (4/7)(10.75/32)^{4/3} = 0.13345$ gives $100\theta_*^{\text{LoN}} = 1.04123$, matching the Planck raw acoustic observable to 0.4σ . No fitting is performed.

The reconstructed Hubble parameter $H_0^{\text{LoN}} = 70.71 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is neither the CMB inference value (67.4) nor the local-ladder value (73.17). Both of those are outputs of their respective dictionaries (base- Λ CDM and Cepheid–SN Ia calibration). The residual discrepancy with the SH0ES value corresponds to a 0.074 mag calibration offset, well within known systematic uncertainties.

The correct statement is therefore not that LoNalogy “solves the Hubble tension,” but rather:

LoNalogy redefines the Hubble problem as an acoustic-dictionary closure problem, and closes it.

The remaining tasks are observational, not theoretical: fitting the Planck raw $TT/TE/EE$ likelihood with the LoNalogy parameterisation, and pulling back the local-ladder raw anchor data into LoNalogy cosmography. These are reductions of existing data under a new dictionary, not extensions of the theory.

Acknowledgements

The author thanks Claude (Anthropic) for assistance with derivation verification, numerical computation, and manuscript preparation. Numerical results were verified using Python/NumPy/SciPy/mp

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